

RESEARCH ARTICLE

How to Mitigate Misinformation Diffusion on Blockchain-Based Decentralized Social Network Platforms? Insights From an Agent-Based Simulation Model

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ABSTRACT

Although the spread of misinformation on centralized social media platforms has received significant attention, few studies compare centralized and decentralized platforms, and how to mitigate misinformation diffusion in newly emerging blockchain-based decentralized social network platforms. We study misinformation diffusion between the decentralized and the centralized platforms and identify three decentralized governance mechanisms to mitigate the spread of misinformation in decentralized networks: user flagging, user article ratings, and user reputation. Our empirical experiments using agent-based simulations leveraging real-world data from two platforms reveal two findings. First, comparing misinformation diffusion between the decentralized Steemit and the centralized Pokec platforms suggests that in the absence of any mitigation mechanisms, misinformation in decentralized platforms affects more users, at faster rates, and reaches shorter distance from the misinformation initiating user. Second, within the decentralized platforms, misinformation affects fewer users, at slower rates, and reaches shorter distance from the misinformation initiating user in the presence of mitigating mechanisms than in their absence. We discuss the implications of these results both for the understanding of misinformation diffusion and for the governance of decentralized social network platforms.

1 | Introduction

The spread of misinformation on social network platforms is a well-known problem with significant consequences for platform owners and businesses. Eli Lilly, for example, suffered a 4.37% drop in stock price due to fake news related to its insulin product (Lee 2022). The increasing use of blockchain technology, coupled with changes such as user propensity to favor decentralized control and concerns about privacy, has led to an interest in blockchain-based decentralized social network platforms (hereafter, decentralized platforms) such as Steemit and Hive (Lacity

et al. 2023; Lumineau et al. 2022). Such decentralized platforms bring together the technological innovation of blockchain and a decentralized business model (Sunar and Swaminathan 2022). Decentralized social network platforms are applications (dApps) that operate on peer-to-peer networks of decentralized autonomous organizations (DAO), a new form of community-based organizing. Users participate in the governance of the platform and avoid control from any central authority (see Figure 1).

Although extant research within the operations management (OM) literature has documented mechanisms for the spread of

Summary

- In the absence of any mitigation mechanisms, misinformation diffusion in the decentralized platforms (relative to centralized platform) affects a larger number of users, at a faster rate, and reaches a shorter distance from the misinformation initiating user.
- Comparing the presence versus absence of the decentralized governance mechanisms within the blockchain-enabled decentralized platforms, misinformation affects fewer users, at slower rates, and reaches a shorter distance from the misinformation initiating user in the presence of user flagging, user article rating, and user reputation.
- Organizations should consider different diffusion dynamics in the decentralized platform to redesign and transform their operational processes for use of social network in engaging with customers and partners.

information (Goel et al. 2016; Yoo et al. 2016); exploitation of the network for benevolent gains such as humanitarian operations (Swaminathan 2018) as well as malicious gain such as fake news (Kitchens et al. 2020); and mitigation mechanisms to stop the spread of misinformation (George et al. 2021), most of this work is in the context of conventional, centralized

social networks (Huang and Mithas 2024; Parker et al. 2016) such as Twitter/X and Weibo. In centralized platforms, the platform owner controls the data, infrastructure, revenue, governance rules for managing complementors and sharing economic benefits, and algorithms to influence the spread of information, including centralized governance mechanisms such as user bans and fact-checking to mitigate misinformation diffusion. Despite the rise and importance of decentralized platforms, few studies have so far compared centralized and decentralized platforms, and how to mitigate misinformation diffusion in newly emerging blockchain-based decentralized social network platforms.

Against this backdrop, we pose two related but distinct research questions: (1) How are misinformation diffusion dynamics different in decentralized platforms in comparison to centralized platforms? (2) How do misinformation mitigation mechanisms differ in their effectiveness to stop the spread of misinformation on decentralized social network platforms? These are important questions because misinformation impacts firm operations (Cezar et al. 2020; Mudambi et al. 2024) and platform governance (Candogan and Drakopoulos 2020; Papanastasiou 2020).

We compare diffusion dynamics between a decentralized (Steemit) platform built on blockchain technology where users' actions and transactions are immutable, users can earn, buy,

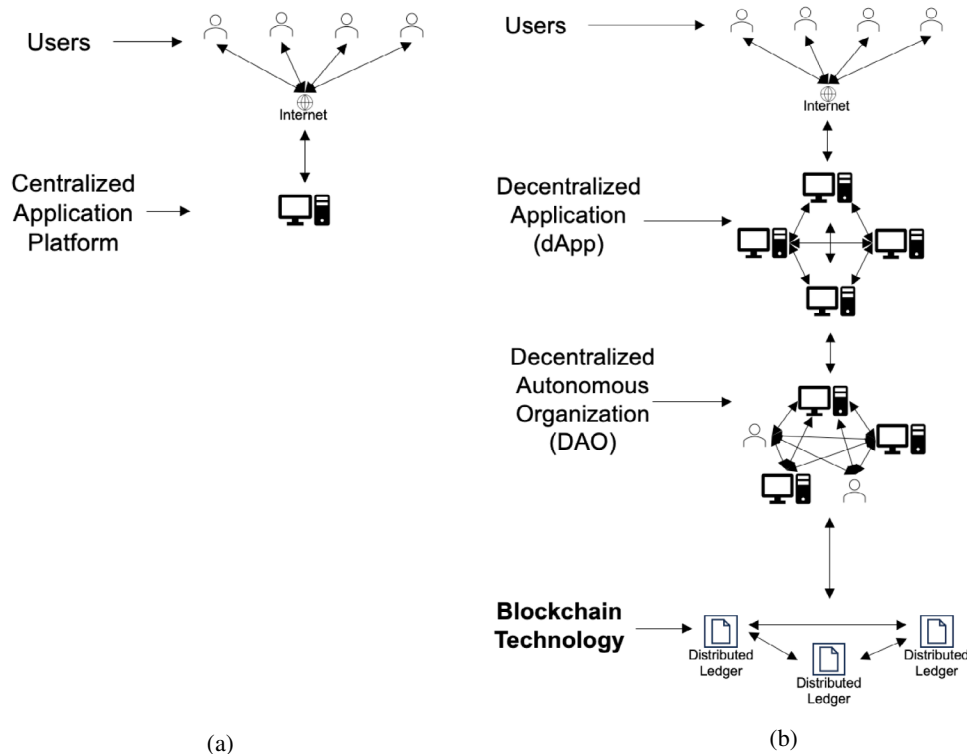


FIGURE 1 | Centralized versus blockchain-based decentralized social network platforms. (a) Centralized social network platform. (b) Blockchain-based decentralized social network platform. A centralized application platform is developed, deployed, and governed by a single business entity (a global controller) that is accessed by users using different modes, such as smartphones or web browsers, via the internet. In the case of decentralized platforms, the blockchain technology underpins the platform and allows for the maintenance of a distributed and peer-to-peer ledger of all transactions and content. Using the blockchain technology, the decentralized autonomous organization (DAO) allows community members to manage and automate transactions related to the DAO, such as financial and social interactions. Rules are codified, copied across all peers in the network, and uniformly used by community members. On top of DAOs, decentralized applications (dApps) are developed that are distributed across multiple participants. Some examples of dApps are Steemit and Hive.

sell, and exchange native tokens for fiat currency, and the platform codifies governance via a decentralized community-based consensus algorithm, and a centralized (Pokec) platform.¹ We also identify and operationalize three decentralized governance mechanisms for misinformation mitigation (user flagging, user article rating, and user reputation) that leverage community consensus (e.g., ratings) to assess the efficacy of these misinformation mitigation mechanisms leveraging real data from the decentralized Steemit platform. To account for the complexity of misinformation diffusion and mitigation dynamics, we develop an agent-based simulation model (ABM) to “capture the rich, nonlinear, and dynamic features typical of operational environments” (Chandrasekaran et al. 2018, 1).

Our simulation experiments show that in the absence of any mitigation mechanisms, misinformation reaches a shorter² distance from the misinformation initiating user, affects a large number of users, and diffuses faster in decentralized platforms compared to centralized platforms. Comparing the presence versus absence of the decentralized governance mechanisms within the blockchain-enabled decentralized platforms, we find that misinformation affects fewer users, at slower rates, and reaches a shorter distance from the misinformation initiating user in the presence of user flagging, user article rating, and user reputation.

Among our key contributions, this is one of the first studies in OM to focus on misinformation on decentralized platforms, in which governance rules are based on community consensus, in contrast to prior work on misinformation in OM which mostly focuses on centralized platforms in which a global controller sets governance rules. Our work also responds to a recent call for OM research to design a “lever that a platform can use to induce desired behavior in social networks” (Candogan and Drakopoulos 2020, 497) by addressing the mitigation of misinformation diffusion with a focus on decentralized platforms. Our ABM-based approach complements prior work in OM (Candogan and Drakopoulos 2020; Mostagir and Siderius 2023) that has considered misinformation diffusion and mitigation dynamics using primarily analytical models (Papanastasiou 2020), and secondary data with econometric models (Mudambi et al. 2024).

2 | Theoretical Framework

2.1 | Background

We draw on two research areas: (1) misinformation mitigation mechanisms for platforms (see Table A1), and (2) governance of blockchain-based decentralized platforms.

First, prior OM studies have considered several factors (e.g., community influence, lack of intervention) that facilitate misinformation diffusion on centralized social network platforms (Candogan and Drakopoulos 2020; Mostagir and Siderius 2023; Papanastasiou 2020). Among misinformation mitigation mechanisms, prior work focuses on the level of impact (account or content) and algorithmic and/or platform-based approaches. Regarding the impact level (e.g., account or content), an account-level focus is to ban a focal agent or limit the set of actions

available to the focal agent, either temporarily or permanently. This approach, although effective at stopping repeat offenders (Ng et al. 2021), has received criticism for potential bias and challenges surrounding the platform's policy to issue such bans. Mitigation mechanisms at the content level include flagging potential misinformation either using experts (Kim et al. 2019) or through signals based on user reputation and content quality (Kim et al. 2019; Ng et al. 2021).

Turning to the algorithmic and/or platform-based approach, algorithmic detection of misinformation (Ng et al. 2023) combines cues from account- and content-levels to flag misinformation (Musmar 2021), while the platform-based approach may implement a shadow ban, in which the visibility of the content is programmatically limited and/or mentions of the focal content may be suppressed (Mudambi et al. 2024). While this body of work informs misinformation mitigation efforts in centralized platforms, there is limited research on the misinformation mitigation approaches that are applicable and effective on decentralized platforms.

Second, in contrast to centralized platforms, governance in decentralized platforms is enforced by means of the underlying blockchain-based network (Lumineau et al. 2021). Community members possess and/or transact to acquire governance tokens that afford voting rights proportionate to tokens' weights (Tsoukalas and Falk 2020). Governance is shared within the community via proposals that facilitate community participation (e.g., via voting). Platform transactions (e.g., post, comment, vote) are codifiable and verifiable using the underlying stable blockchain network (Ilk et al. 2021). Consensus rules are codified and take the form of formal computer code, known as smart contracts (Zhao et al. 2022), thereby limiting outsized influence by specific users. Decision rights to resources are managed using the blockchain, and incentives are in place for peer-to-peer consensus (Beck et al. 2018). Communities often define custom rules (e.g., transaction fees) that govern the platform (Shang et al. 2023).

Building on these governance properties of decentralized platforms, the verifiability property of blockchain is used to identify sources of misinformation (Petratos and Faccia 2023). Also, blockchain offers autonomous enforceability, which reduces dependency on law (contractual governance) or future transaction value (relational governance) using codified rules that are based on community consensus. Although this body of work offers insights on the operational and governance aspects of decentralized platforms, only limited work addresses the governance challenge of managing the spread of misinformation on decentralized platforms, which we address in this research.

2.2 | Complex Adaptive Systems Perspective on Platforms

We draw on the complex adaptive systems (CAS) perspective (Choi et al. 2001; Nair and Reed-Tsochas 2019) to highlight similarities and differences related to control and governance rules between centralized and decentralized platforms. The CAS perspective provides a flexible lens for modeling

complex real-world phenomena, such as networks of agents (Nair and Reed-Tsochas 2019), and the behavior of users and their interdependence, which may result in emergent and unpredictable outcomes in both centralized and decentralized platforms.

Both centralized and decentralized platforms can be construed as complex adaptive systems, because they exhibit the following four complexity properties (Choi et al. 2001; Nair and Reed-Tsochas 2019). First, heterogeneous platform users are neither too similar nor too dissimilar with respect to their participation objectives, preferences, and constraints. Second, users form connections (e.g., friend, follower) that may differ along the dimensions of temporality (the addition of new connections over time), frequency (the number of connections in a period), content (the types of activities), and directness or indirectness. Third, users are interdependent (two connected users gain value) because the utility for each user is directly related to the availability and participation of other users on the platform. Also, interdependence between users is asymmetric—a focal user may be highly dependent on another connected user, while the connected user's dependence on the focal user is relatively low. Finally, asymmetric interdependence between connected users can lead to nonlinearities and unpredictable outcomes in the system. In a nonlinear system, two comparable initiating events (e.g., the initiation of misinformation) may exhibit different outcomes (e.g., number of users affected by misinformation) on the platform.

Two properties distinguish decentralized from centralized platforms: the lack of global control and the allocation of economic benefits based on rules defined by community consensus. First, in the case of centralized platforms, global control is exerted by the platform owner, who defines rules and governance mechanisms and retains control over access to the platform data. Thus, the global controller can implement an array of mechanisms such as limiting broadcasting information, facilitating special interest groups, boosting specific content based on the initiator, content type, or geographic attributes, and employing proprietary algorithms to alter users' behavior and information diffusion. The presence of a global controller leads to control and arbitrary changes to rules. For example, the social network platform X prohibits the inclusion of links to competing platforms (Gerken 2022), and the social network platform Sina Weibo banned 1120 accounts that criticized the government's policies on the COVID-19 outbreak (AP 2023). In contrast, decentralized platforms rely instead on decentralized community-based mechanisms to define governance rules that are different from those in centralized platforms. For example, X does not limit the number of upvotes/likes given by a user, whereas the consensus-based algorithm on the decentralized platform Steemit limits the number and value of upvotes offered by a user within a period.³

Second, centralized and decentralized platforms also differ in the allocation of economic benefits.⁴ Users' participation on centralized and decentralized platforms is driven by motives that are utilitarian (e.g., seeking relevant information, such as news or transactions), hedonic (e.g., connecting and interacting with others), and economic (Choi et al. 2020; Delkhosh et al. 2023; Kim et al. 2019). Economic motives relate to

financial benefits received by participating users. For example, users on centralized platforms might receive payments from an advertiser to promote a product, or users on decentralized platforms might, in exchange for curation contributions (e.g., vote), receive native tokens that can be exchanged for fiat currency.

On a centralized platform, the global controller determines the rules for allocating economic benefits without the agreement of participants, and there is no mechanism to seek consensus and/or feedback from the community on the potential implications of any rule changes. In contrast, decentralized platforms are subject to consensus-based governance—rule establishment and changes. Each community member can contribute toward the proposed rule (e.g., vote on acceptance or rejection), and changes require community consensus. Importantly, since the rules are codified using the underlying blockchain in decentralized platforms, participating members are obligated to adopt any such rule changes. Consequently, no single community member can alter rules, transactions, or other platform actions.

Differences in allocation mechanisms on centralized versus decentralized platforms lead to different allocation rules. For example, community-based value assigned to focal content on the centralized Reddit platform is associated with the content poster (e.g., upvotes by the community translate to platform-specific tokens). In contrast, community members on the decentralized platform Steemit vote on the payout structure for users who have authored and curated content; currently, 75% of the tokens are paid to the content author, while 25% of the tokens are unequally distributed among curating users.

2.2.1 | Misinformation Diffusion on Centralized and Decentralized Platforms

Based on the discussion above, we posit that misinformation diffusion dynamics are different on decentralized platforms compared to centralized platforms for three reasons. First, the global controller on a centralized platform dictates the diffusion dynamics (e.g., limits the number of connection requests, stops the sharing of content, deletes content, or introduces a recommender system). In contrast, the lack of a global controller on a decentralized platform implies the existence of organic diffusion dynamics based on the social network structure within the constraints set by community-based consensus (e.g., the number of votes in a period).

Second, the global controller-based allocation of benefits on the centralized platform implies an unpredictable, mutable, and discriminatory reward system, if any (e.g., a celebrity user may receive a higher reward than a noncelebrity user for comparable content), whereas, the community-based allocation of economic benefits on the decentralized platform implies predictable, immutable, and equal rewards for community-valued contributions (e.g., all content authors receive 75% of the associated payout). Differences related to the allocation of economic benefits influence users' contribution behavior. For example, Maleki et al. (2023) find that health-related content on decentralized platforms is relatively positive, joyful, analytical, informational,

of better quality, and clickbait, and that users are more likely to like and comment on a decentralized platform in comparison to a centralized platform.

Finally, unlike most centralized platforms, users on a decentralized platform receive rewards for community-valued original contributions and curation efforts. For instance, users on Steemit receive a reward to upvote content, with larger rewards for earlier curations. Consequently, users are incentivized to develop greater first-degree connections on the platform (e.g., get notifications of new content on the platform).

In summary, our key argument is that the centralized mechanisms in which a global controller plays an active role may not be applicable for decentralized platforms because misinformation mitigation mechanisms must align with the decentralized decision-making processes of decentralized platforms. Also, the conventional approach of punishment (e.g., banning a user) is not applicable for decentralized platforms, since users cannot be banned from accessing and transacting over the blockchain because of the blockchain-based immutability of records, such that deleting or locking content is impermissible.

2.2.2 | Decentralized Governance Mechanisms for Misinformation Mitigation in Decentralized Platforms

We draw on commons-based governance theory (Ostrom 1990), which has its origin in the management of self-governed common-pool shared community resources such as forests, rivers, and fisheries, among others. The theory provides eight organizing principles that are found in successfully governed communities and have been applied to online environments in the study of the governance of communities that manage shared resources, such as Wikipedia (Forte et al. 2009). Commons-based governance is rooted in the community-based approach where governance is based on rules that are defined and enforced using community consensus. In our context, the managed resource is the focal content that propagates through the network.

We focus on a non-exhaustive set of decentralized misinformation mitigation mechanisms: user flagging, user article rating, and user reputation.⁵ These mechanisms are drawn from mitigation approaches considered in prior work focusing on centralized platforms: the user flagging mechanism draws from flags provided by users (Ng et al. 2021); the user rating mechanism draws from ratings provided by users (Kim et al. 2019); and the user reputation mechanism draws from users' ratings about the content author (Kim et al. 2019). We label these mechanisms as decentralized governance mechanisms because of their emphasis on community participation for identifying and mitigating the spread of misinformation.⁶

We consider the influence of decentralized governance mitigation mechanisms at the platform level, focusing on explicit mechanisms—that is, they offer tangible, comparable, and observable signals to the entire community.⁷ The conceptualization and operationalization of our chosen mechanisms are based on content that is factually incorrect (misinformation) rather than factually correct but perceived to be of lower quality. However, we acknowledge that the perception associated with

content is such that the content may be credible (e.g., receive a high rating on the platform) while being factually incorrect, or it may be untrustworthy (e.g., posted by a user with a relatively low reputation score) while being factually correct.

3 | Methods

3.1 | Data

To compare misinformation diffusion between centralized and decentralized platforms, we use publicly available data from the centralized social network platform Pokec (Takac and Zabovsky 2012).⁸ Users' actions include original contributions (making an original post related to a topic), curation contributions (commenting), and interpersonal connections (such as friending or direct messaging). Data related to users' activities are retained by the platform owner(s), and the platform's revenue is derived predominantly from advertisement.

The Pokec dataset consists of 1,632,803 users who registered on the platform from its inception in November 1999 through May 2011. In Pokec, not all user contributions receive economic benefits from the platform; popular users may be paid by marketers to post content related to a focal product and/or service. Allocation of economic benefits, if any, is determined by the platform, and these benefits are shared predominantly via fiat currency. The users' view of the content is influenced by the global controller (e.g., through a recommender system). Pokec does not support community-driven valuation of content. A user's attribute (e.g., the user's number of friends) cannot be exchanged for fiat currency.

To compare the presence versus absence of the decentralized governance mechanisms within the blockchain-enabled decentralized platforms, we use publicly available data from the decentralized social network platform Steemit (Li et al. 2021). Steemit is built using the STEEM blockchain, where data on content, user activity, and transactions are stored on the blockchain, making the data both publicly accessible and immutable. The dataset comprises 1,316,499 users who registered on the platform from its inception in March 2016 through November 2019. Steemit operates on the proof-of-brain principle, which focuses on the community-based assessment of content value. This is unlike the proof-of-work principle (in which members compete to update the blockchain, which leads to rewards), used by popular cryptocurrencies such as Bitcoin, which emphasizes community-based verification of work and requires computational resources, in contrast to the proof-of-brain approach that emphasizes non-computational resources (e.g., time and effort). Content contributors (e.g., the authors) and curation contributors (e.g., the upvoters) receive tokens for their contributions.

In Steemit, rules related to the allocation of benefits for contributions are established and voted upon by the community rather than defined by a global controller. Platform participation is driven by different objectives (e.g., exploration of social media), motivations (e.g., earning tokens), constraints (e.g., mode of access), and preferences (e.g., type of content). Users' actions include authoring content, curating, and developing interpersonal connections (e.g., following other users). In addition to tokens

TABLE 1 | Descriptive statistics for data and empirical context in the agent-based simulation model.

Network property	Centralized social network platform (Pokec)	Blockchain-based decentralized social network platform (Steemit)
Number of <i>nodes</i> (agents/users)	1,316,499	1,316,499
Number of <i>edges</i> (directed connection between nodes)	29,259,411	115,645,108
Average <i>degree</i> (number of nodes connected to the focal node)	66.479	346,100.94
<i>Density</i> —proportion of actual to possible connections in the network (value between 0 and 1). Greater value (closer to 1) represents a dense network where information flows easily relative to a sparse network (density closer to 0)	1.688×10^{-5}	6.672×10^{-5}
<i>Reciprocity</i> —likelihood of a symmetric relationship between nodes. Values range from 0 to 1; values closer to 0 (1) indicate a greater (smaller) likelihood of asymmetric relationship	0.548	0.186
<i>Edge</i> (connection between nodes)	Directional (receives updates related to actions on the platform)	Directional (receives updates related to actions on the platform)
<i>Network portrait divergence</i> —measure that quantifies dissimilarity between two networks. Values range from 0 to 1; values closer to 0 (1) indicate similar (highly dissimilar) networks		0.6821
<i>Underlying platform/system</i>	Proprietary	STEEM blockchain

Note: Based on the global statistics and the network portrait divergence measure, we conclude that centralized (Pokec) and decentralized (Steemit) networks are structurally dissimilar. Also, the degree distribution of our data samples shows a better fit for the power law distribution, which empirically supports our conceptualization of centralized and decentralized platforms as complex adaptive systems. For Pokec, the data span from November 1999 through May 2011 (Takac and Zabovsky 2012). For Steemit, the data span from March 2016 through N 2019 (Li et al. 2021).

received for community-valued contributions, each user has an associated reputation score, which is directly influenced by contribution quality (for both original and curation contributions). Finally, users transact tokens to earn fiat currency. The value of STEEM arises from limited supply and trading on crypto exchanges. A focal user's STEEM balance does not enable her to influence others' contributions.⁹

Table 1 provides descriptive statistics on Pokec and Steemit dataset.

3.2 | Agent-Based Model

We develop an agent-based simulation model (ABM) to model the spread of misinformation and evaluate decentralized misinformation mitigation mechanisms on decentralized platforms.

3.2.1 | Modeling the Spread of Misinformation on Centralized Versus Decentralized Platforms

Building on prior work (Goel et al. 2016; Yoo et al. 2016), we model the influence of the consensus algorithm for the allocation of economic benefits in our decentralized model as follows. In line with the community-based allocation of economic benefits, accurate and early adoptions of highly valued content (e.g.,

upvotes, comments) receive relatively larger economic rewards. However, the value of the content is often realized over time as more users adopt that content, which requires a contributor to decide whether adopting that focal content aligns with her objectives. Equation (1) describes the user reward for contribution (Gonzalez 2017; Steem 2017):

$$\text{Reward to contributor} = \frac{\sqrt{RS_1} - \sqrt{RS_0}}{\sqrt{RS_T}} \quad (1)$$

where RS_0 is the accumulated tokens (e.g., the total weight of votes) associated with the post before the focal contribution; RS_1 is the tokens associated with the focal contribution (e.g., the weight of the upvote), and RS_T is the total tokens associated with the post before the payout is computed. For a focal user, when the expected payout is larger than her threshold (drawn from a uniform distribution), we allow content adoption via external influence. This approach models the relatively larger rewards received for earlier adoptions while allowing users to probabilistically reject adoptions.

Following prior work (Goel et al. 2016; Törnberg 2018; Vosoughi et al. 2018), we use three measures to evaluate the efficacy of misinformation mitigation mechanisms while controlling for differences in network size. First, *structural virality* is the average depth of users in the diffusion tree (Goel

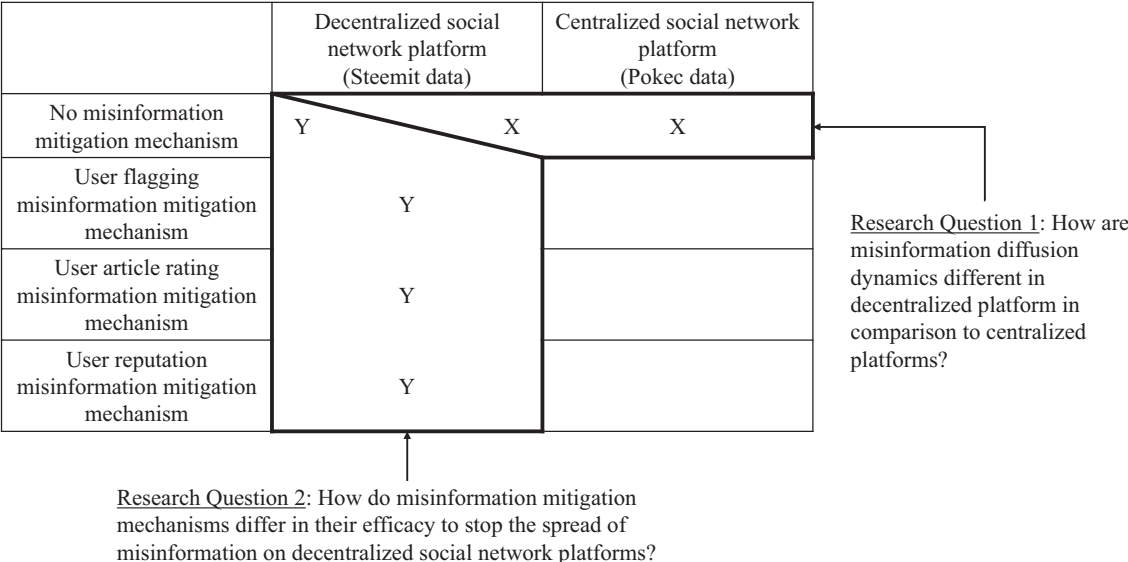


FIGURE 2 | Research focus and empirical experiments. Notes: This figure maps our research focus and key empirical experiments. For Research Question 1, we compare Xs in the figure between centralized and decentralized platforms that use Pokec and Steemit datasets, respectively. For Research Question 2, we compare Ys in the figure within decentralized platforms using four scenarios: no decentralized governance mechanisms, user flag, user article rating, and user reputation.

et al. 2016). The measure interpolates between broadcast and viral diffusion. For broadcast (a relatively low value of structural virality), just a few users are responsible for the majority of propagation, whereas for viral diffusion, we observe a multigenerational branching process. Second, *diffusion size* is the ratio of the number of users in the diffusion tree to the number of users in the tree as a whole (Vosoughi et al. 2018). The whole tree consists of the descendants of a focal user initiating a cascade, whereas the diffusion tree is a subset of the tree in which all users have adopted the information. Finally, *global cascade* is the number of periods required for a diffusion tree to reach 50% diffusion size (Törnberg 2018). The goal for the misinformation mitigation mechanism is to reduce structural virality and diffusion size, but increase the time to reach 50% diffusion size.

3.2.2 | Modeling Decentralized Governance Mechanisms for Misinformation Mitigation in Decentralized Platforms

We model three decentralized governance mechanisms for misinformation mitigation: user flagging, user article rating, and user reputation. In the case of user flagging, platform users flag content as misinformation. For a focal content, the number of users who have flagged the content is visible to all users. In the case of user article rating, users provide ratings for the focal content. For each focal content, the average rating (e.g., an average rating of 4.2 out of 5) is visible to all users of the platform. Content having a relatively lower rating is categorized as misinformation/inaccurate, whereas content with a relatively high rating is categorized as community-valued content. Finally, user reputation rating uses the author's reputation as a proxy for the focal contribution's rating to the platform. Consequently, the author's reputation represents a content-related signal to the community.

To design decentralized governance mechanisms for misinformation mitigation, we consider the actual content diffused in the network (relevant for user flagging and user article rating), as well as the reputational signal of the cascade initiator relevant for user reputation rating. Conceptually, each mechanism provides a signal to a focal user, whose probability of adopting the focal misinformation decreases with the strength of the mechanism.

Figure 2 maps our research question and empirical experiments.

4 | Results

4.1 | Misinformation Diffusion on Centralized Versus Decentralized Platforms

Figure 3 contextualizes the misinformation diffusion dynamics of centralized versus decentralized platforms to show that relative to centralized platforms, misinformation diffusion on decentralized platforms has lower structural virality, greater diffusion size, and requires fewer periods to reach global cascade.¹⁰

Figure A1 shows the results of the analysis examining the difference in misinformation propagation in centralized versus decentralized platforms.¹¹ We note three points. First, we note the relatively lesser influence of neighbors (Figure A1a) and greater number of neighbors (Figure A1b) in decentralized platforms. As the number of users that can influence a focal user to adopt information increases, the influence of each individual neighbor decreases. Consequently, a focal user realizes greater affordances for adoption. Second, since the average depth of the diffusion tree for centralized platforms is

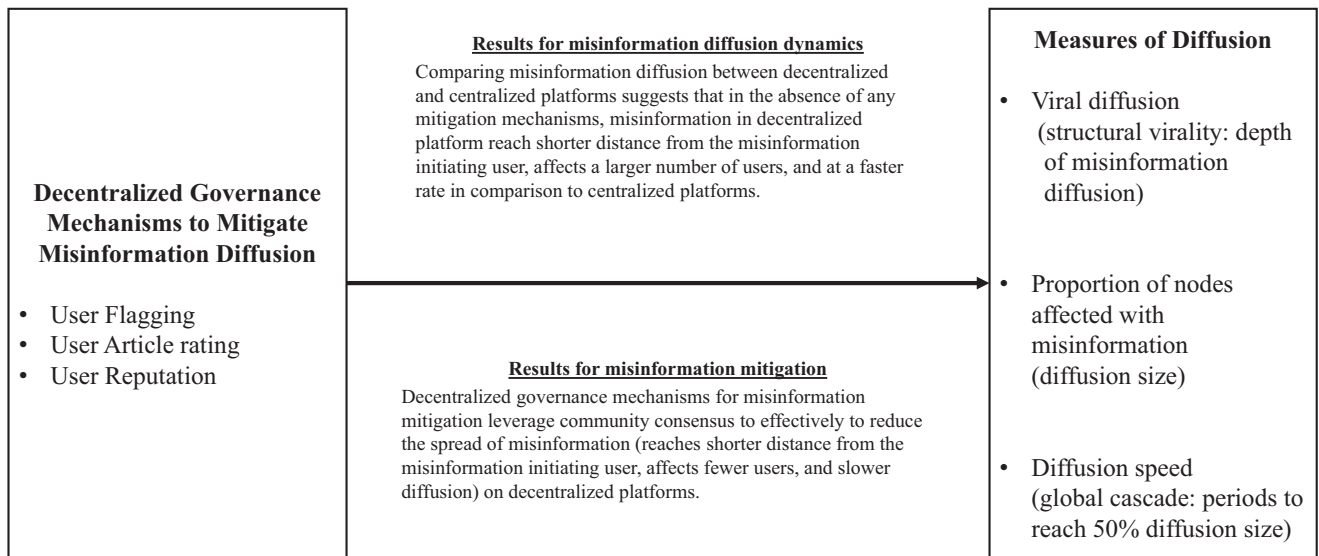


FIGURE 3 | Research context and empirical results. Notes: This figure contextualizes our study results. Results for misinformation diffusion dynamics show that in the absence of any mitigation mechanisms, misinformation in decentralized platforms reaches shorter distance from the misinformation initiating user, affects a larger number of users, and at a faster rate in comparison to centralized platforms. Results for misinformation mitigation suggest that decentralized governance mechanisms for misinformation mitigation (user flagging, user article rating, and user reputation) are effective in reducing the spread of misinformation on decentralized platforms.

larger than that for decentralized platforms (Figure A1c), we observe relatively greater structural virality in the centralized platforms. Finally, Figure A1d plots the average ratio of adopted neighbors for each period. We note a relatively greater proportion of adopted neighbors. Consequently, fewer periods are required to reach 50% diffusion size on decentralized platforms.

4.2 | Efficacy of Misinformation Mitigation Mechanisms in Decentralized Platforms

We evaluate the efficacy of misinformation mitigation mechanisms in the decentralized platform using data from the Steemit platform. Our results suggest that decentralized governance mechanisms are effective in reducing the spread of misinformation on decentralized platforms (permeates shorter distance from misinformation initiating user, fewer users are affected, and slower diffusion of misinformation). This is because relative to a decentralized platform with no mitigation mechanism, we observe lower structural virality, lower diffusion size, and greater time required to reach a global cascade in systems characterized by misinformation mitigation mechanisms.

Figure A2 shows the results of an analysis investigating the impact of different misinformation mitigation mechanisms on decentralized platforms.¹² We note three points. First, focusing on the relative efficacy of misinformation mitigation mechanisms compared to decentralized platforms without any mechanisms, Figure A2a shows that the proportion of adopted neighbors is consistently higher in decentralized platforms without any mechanisms. Since the proportion of users that adopt via external influence is comparable across all systems, Figure A2a suggests that the proportion of users that adopt via internal influence is higher in decentralized platforms without

any misinformation mitigation mechanism, thus providing evidence of the efficacy of such mechanisms in reducing the spread of misinformation. Second, Figure A2b shows that the user flagging system has the highest proportion of users impacted by the mitigation mechanism. Hence, we observe lower diffusion size in the user flagging mechanism system. Finally, Figure A2b also shows that the proportion of users impacted is greater in platforms utilizing user-reputation mitigation mechanisms compared to those employing user-article-rating mechanisms. In the case of decentralized platforms, users' contributions directly impact their reputation score; community-based consensus mechanisms incentivize contributions that are valued by the community and platform.

Table A2 summarizes our empirical analyses.¹³

5 | Discussion

5.1 | Findings and Contributions

Based on extensive ABM experiments, we find that misinformation diffusion in blockchain-based decentralized platforms reach shorter distance from the misinformation initiating user, affects a larger number of users, and at a faster rate in comparison to centralized platforms. Also, we find that commons-based misinformation mitigation mechanisms effectively manage the spread of misinformation on blockchain-based decentralized platforms such that misinformation affects fewer users, at slower rates, and permeates shorter distance from the misinformation initiating user in the presence of mitigating mechanisms than in their absence.

Our empirical findings complement and extend prior work on misinformation mitigation in several ways. First, prior work has proposed strategically placing seed users (e.g., bots) in the

network that stop the spread of misinformation and/or introduce correct information (Ghoshal et al. 2021; Yoo et al. 2016). Such an approach requires learning the community structures (e.g., the strongly connected users) and having the ability to place seed users, which is unavailable in decentralized networks. Nevertheless, the mechanisms considered in our work align with this stream of research based on the strategic and differing economic benefits in decentralized platforms. Similarly, prior work also recognizes the use of community structures to learn prominent users (those occupying a position that connects subcommunities). Our framework accounts for users' prominence (i.e., influence) through their reputation: for example, a curation rating by a highly reputed user—one who is connected to multiple communities—is highly valued by the community.

Second, our conceptualization of decentralized platforms as a complex system extends the complex adaptive systems perspective adopted in the OM research. Specifically, OM research has employed the complex adaptive systems perspective to model complex adaptive operational and supply chain systems in which agents are firms that interact, adapt, and coevolve with changes to their local and global environments (Choi et al. 2001; Nair and Reed-Tsochas 2019). Such a perspective has led to investigations of operational issues within focal complex systems, such as improving system-level transaction success in surplus-driven supply networks (Dhanorkar et al. 2019) and the reduction of the negative effects of supply chain network disruptions (Zhao et al. 2019), among others. We extend this perspective by modeling individual users of a decentralized platform to address the governance challenge of mitigating misinformation diffusion and showing that macro-level system differences (e.g., the presence/absence of a global controller and the community-based allocation of economic benefits) lead to different system-level outcomes (e.g., differences in misinformation diffusion dynamics and efficacy of misinformation mitigation mechanisms).

Third, we extend the theory on governance of blockchain-based decentralized platforms. Specifically, our commons-based governance mechanisms complement existing governance mechanisms on decentralized platforms that are based on rewards (Belavina et al. 2020), tokens (Gan et al. 2023), and smart contracts (Cong and He 2019). Further, our conceptualization and operationalization of commons-based misinformation mitigation mechanisms extend the application of commons-based governance principles (Ostrom 1990) to blockchain-based community-governed decentralized platforms.

Finally, our work aligns with studies focusing on the operational consequences of agents' self-organization leading to consensus on information (e.g., resources available during a natural disaster) and/or action (e.g., key areas of focus in response to a natural disaster) (Nan and Lu 2014; Sunar and Swaminathan 2022). However, our study extends this stream of research by focusing on agents' consensus for purposes of governance (e.g., to mitigate misinformation) rather than collective action (e.g., to respond to a disaster). Also, prior work considers agents' self-organization in social network settings where a global controller dictates the rules of engagement (e.g., settings where there is a proprietary platform supporting user-generated content). In

our case, we consider decentralized platforms, in which a global controller cannot set the rules of engagement.

5.2 | Implications for Research

Our work provides three implications for the nascent literature on operational applications of blockchain technology and the use of blockchain-based decentralized social network platforms for OM. First, this study evaluates the efficacy of community-driven governance mechanisms to mitigate the spread of misinformation on decentralized social network platforms. Although OM research recognizes the potential of blockchain technology to address malicious activity (e.g., counterfeits (Pun et al. 2021)) and highlights the need to consider the effects of misinformation using blockchain technology (Petratos and Faccia 2023), there is a lack of research that considers data-driven governance mechanisms to mitigate spread of misinformation on decentralized platforms. A key implication of our study is for CAS conceptualization in OM research, which thus far has focused on the microlevel adaptations by agents (e.g., organizations) within the environment, both perceived and external, that give rise to macro-level emergent system properties (Nair and Reed-Tsochas 2019). Our study extends Nair and Reed-Tsochas's (2019) framework of complex adaptive systems in OM research by highlighting the impact of community-based rules on the outcomes of complex adaptive systems.

Second, we explain how and why misinformation diffusion in a blockchain-driven decentralized platform is likely to be different from that in a centralized platform, contributing to the nascent literature on blockchain technology and its applications for operational issues (Babich and Hilary 2020; Choi et al. 2020; Lumineau et al. 2022). Specifically, the current efficiency-based view of OM research on blockchain technology focuses on blockchain-driven solutions to improve information and decision-making in supply chain and associated risk management (Babich and Hilary 2020). Our research advances a complementary, behavior-based view for OM research in which the focus is on highlighting changes in user behavior on blockchain-driven decentralized platforms. Our findings suggest that users' motivations and behaviors in blockchain-driven applications are different from those in existing advertisement-driven centralized platforms. Consequently, current theory on users' behavior and solution processes for operational challenges on centralized platforms may not be directly applicable to decentralized platforms.

Finally, our findings have implications for prior work on the use of social networks and their data for organizational efficiency (Cui et al. 2018; Gunarathne et al. 2018; Lam et al. 2016). Following the descriptive, prescriptive, and predictive framework (Swaminathan 2018), organizations have leveraged social networks for sensing operational issues, such as recalls (Gao et al. 2022); addressing operational issues in real time, such as customer service (Gunarathne et al. 2018); and proactively managing operational issues, such as customer opinion, by leveraging social networks (Gao et al. 2022; Yang et al. 2019). Our findings introduce the two additional dimensions of diffusion speed and size, and motivations for the type of information relevant for decentralized platforms. Specifically, organizations

adapting the descriptive, prescriptive, and predictive framework to a decentralized platform may need decision support systems that internalize the greater speed and broader diffusion in decentralized platforms.

5.3 | Managerial Implications

Our study has three managerial implications. First, our findings highlight the efficacy of commons-based mechanisms to address the larger and faster misinformation diffusion in decentralized platforms. A direct implication of this result for DAO-based platforms is to adopt and retain commons-based mechanisms, especially in the face of community members introducing proposals to remove or alter commons-based mechanisms (Steemit 2018). Our findings directly inform community proposals that focus on commons-based mechanisms on decentralized platforms. Relatedly, our findings suggest that a confluence of commons-based signals enables users to better assess content quality and is effective in mitigating the spread of misinformation.

Second, this research underscores the importance of weighted tokens used in commons-based mechanisms (Tsoukalas and Falk 2020). Consequently, DAO-based platforms need to design mechanisms that account for collusion among users. For example, connected (e.g., friends, followers) users may reciprocate to upvote content. Relatedly, DAO-based platforms should consider vote abuse (Zhao et al. 2022). For example, downvoting content on the Steemit platform is limited to users with relatively greater reputation scores than the author's reputation score. DAO-based platforms need dynamic approaches to mitigate malicious user (users with relatively greater reputation) activity on legitimate content.

Finally, our results highlight operational opportunities and challenges for organizations on decentralized platforms. For example, on centralized platforms, organizations contract with popular users to diffuse product/service information. These popular users enjoy a price premium (e.g., charge more to post) to diffuse information. However, larger and faster diffusion dynamics on decentralized platforms reduce the popular users' premium, benefiting organizations' diffusion goals. On the other hand, larger and faster diffusion dynamics on decentralized platforms require faster organizational responses (e.g., customer service, customer opinion) and revise the playbook for handling crises using social networks (Gao et al. 2022; Gunaratne et al. 2018; Yang et al. 2019), relative to that offered on centralized platforms.

5.4 | Limitations and Future Research Directions

We note three limitations of our study. First, there may be some concerns about our data sample (use of different time frames and duration of our datasets), the exclusion of users that have been deleted and/or users with no relationships on the platform. However, we believe that the impact of these limitations is restricted because our networks represent organic connections developed by users. We also employ random choice in the determination of the cascade initiator and account for inherent differences in users' motives. Additionally, our results are supported by robustness tests (see [Supporting Information: Ecompanion C](#)) using data from Petster, another centralized platform, the same duration of

datasets, and concurrent consideration of mechanisms, among others.

Second, our results are limited to experiments using data from the Steemit platform. Although there is a need for care in expecting or attributing generalizability in any single study (Mithas, Chen, et al. 2022; Mithas, Xue, et al. 2022; Shadish et al. 2002), we are cautiously optimistic that many of our findings may be generalizable to other contexts subject to verification by other studies. This is because our experiments model diffusion dynamics using the information diffusion theory, providing robust empirical evidence related to differences in community structures between centralized and decentralized networks, and draw on real-world data from large social networks.

Finally, our use of an agent-based model to simulate misinformation diffusion inherits the limitations of that method (e.g., use of distributions for agents' behavioral rules, limited adaptations of agents, model complexity drives uncertainty in the results) (Davis et al. 2007). Within the boundaries of these limitations, our findings are robust to alternate model specifications for misinformation diffusion and robustness tests ([Supporting Information: Ecompanion C](#)), allow us to empirically show the impact of the global controller and the community-based allocation of economic gains on the efficacy of misinformation mitigation mechanisms, and enable us to theorize and empirically document differences in misinformation propagation between centralized and decentralized platforms.

This research suggests three major opportunities for future research. First, we compare the efficacy of three commons-based misinformation mitigation mechanisms on decentralized platforms. In practice, decentralized platforms are likely to deploy a portfolio of mitigation mechanisms that includes both static mechanisms (e.g., contracts) and dynamic mechanisms (e.g., algorithms). Future research might consider the interplay between the mechanisms in such a portfolio and its effect on the efficacy of such systems.

Second, given the decentralized organization of newer platforms, future research might consider blockchain-based smart contracts that augment our algorithmic approach, as these may influence users' payoffs based on their behavior related to information diffusion and mitigation. Such approaches would extend existing blockchain-based solutions that focus on tracking changes to content using a commonly maintained ledger (Lumineau et al. 2022; Zhao et al. 2022).

Finally, future research might consider alternate methodological approaches, such as econometric modeling and field experiments, to consider both micro- and macro-level dynamics on decentralized platforms and consider integrating recent advances such as large language models (LLM) to augment agents' dynamic decision-making and behavioral attributes.

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Endnotes

- ¹ There exists a different class of social networks known as federated social networks (e.g., Mastodon, Bluesky), in which self-hosted services, those in which the service operators retain control of their service, interact to create the social network; these social networks do not use the underlying blockchain technology. Our research specifically studies blockchain-based decentralized platforms and does not consider other types of federated networks.
- ² Consider a network with three nodes (A, B, C) with arcs (A, B), (B, C), and (C, D). If A is the misinformation initiating user, we are considering the average of distance for all users affected by the misinformation. Assuming B and C are affected, the average distance will be $(1+2)/2 = 1.5$. In another case where all users are affected, the average distance will be $(1+2+3)/3 = 2$. In the above two examples, we say that the misinformation permeates shorter distance in the first case (average distance is 1.5) in comparison to the second case (average distance is 2).
- ³ In the case of centralized platform Reddit, subreddit administrators must submit a formal request to the global controller to change subreddit status (e.g., public or private), with implications for community-led governance. A lenient and/or passive global controller is not the same as the lack of a global controller, because a centralized platform's rules and governance mechanisms (even if initial) are defined by the global controller, whereas on a decentralized platform it is the community that defines the governance rules.
- ⁴ Current centralized platforms form a continuum related to economic benefits. For example, users on Hacker News receive no economic benefit for contributions, whereas users on Weibo receive an economic benefit for contributing. We account for this diversity of centralized platforms by limiting our discussion to cases in which the allocation of economic benefits is an inherently distinguishing characteristic. We thank the AE for suggesting this point.
- ⁵ **Supporting Information:** Ecompanion A maps the organizing principles of commons-based governance to decentralized governance misinformation mitigation mechanisms of user flagging, user article rating, and user reputation.
- ⁶ Related labels include commons-based or community-based mechanisms.
- ⁷ **Supporting Information:** Ecompanion A provides additional details related to blockchain-based decentralized platforms, explains the conceptual difference between centralized and decentralized platforms, and maps the organizing principles of commons-based governance to misinformation mitigation mechanisms. In comparison, the conventional approach of norm enforcement (e.g., shunning a specific user) is an implicit mechanism which is subjective (i.e., imperfectly inferred from data). Consequently, the implicit approach requires actual data on community behavior with ground truth (e.g., interviews with community members validating norm enforcement). Further, unlike the platform-level consistency of decentralized governance mechanisms, norms are elastic, localized, and evolve over time, which further contributes to their subjectivity.
- ⁸ The Pokec network is represented as a directed multi graph where each user is a node, and an edge is a unidirectional connection between nodes. Edges (friendship in Pokec) are comparable due to their directionality and purpose. In our networks, a directed edge from $x \rightarrow y$ does not imply existence of edge $y \rightarrow x$.
- ⁹ The Steemit network is represented as a directed multi graph where each user is a node, and an edge is a unidirectional connection between nodes. Edges (followership in Steemit) are comparable due to their directionality and purpose. In our networks, a directed edge from $x \rightarrow y$ does not imply existence of edge $y \rightarrow x$. **Supporting**

Information: Ecompanion B provides additional details related to appropriateness of our data collection and sampling strategy, operationalization of misinformation diffusion and mitigation mechanisms, and the empirical study.

- ¹⁰ We model the influence of a global controller curation algorithms (e.g., a recommender system) deployed by the global controller. Specifically, we allow content adoption via external influence if the relative distance between the focal user's and content author's reputations (using the *PageRank* score, which is computed based on network position) is greater than or equal to a randomly generated threshold drawn at random from a uniform distribution. This approach allows the model to capture the uncertainty associated with a focal user adopting recommended content. In each simulation run, we identify a cascade initiator at random in each system. We use the same parameters (e.g., number of periods, mean threshold for internal adoption) across both systems for comparability. At the end of each simulation run, we log data related to structural virality, diffusion size, and global cascade.
- ¹¹ **Supporting Information:** Ecompanion C provides additional empirical tests to support this result and discusses two additional mechanism tests (impact of incentives on decentralized platforms, and modeled influence of the global controller and the community-based consensus algorithm on centralized and decentralized platforms, respectively).
- ¹² **Supporting Information:** Ecompanion C.4 presents empirical experiments comparing the efficacy of misinformation mitigation mechanisms in centralized platforms.
- ¹³ **Supporting Information:** Ecompanion C presents empirical tests to support our main findings and exploratory findings.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

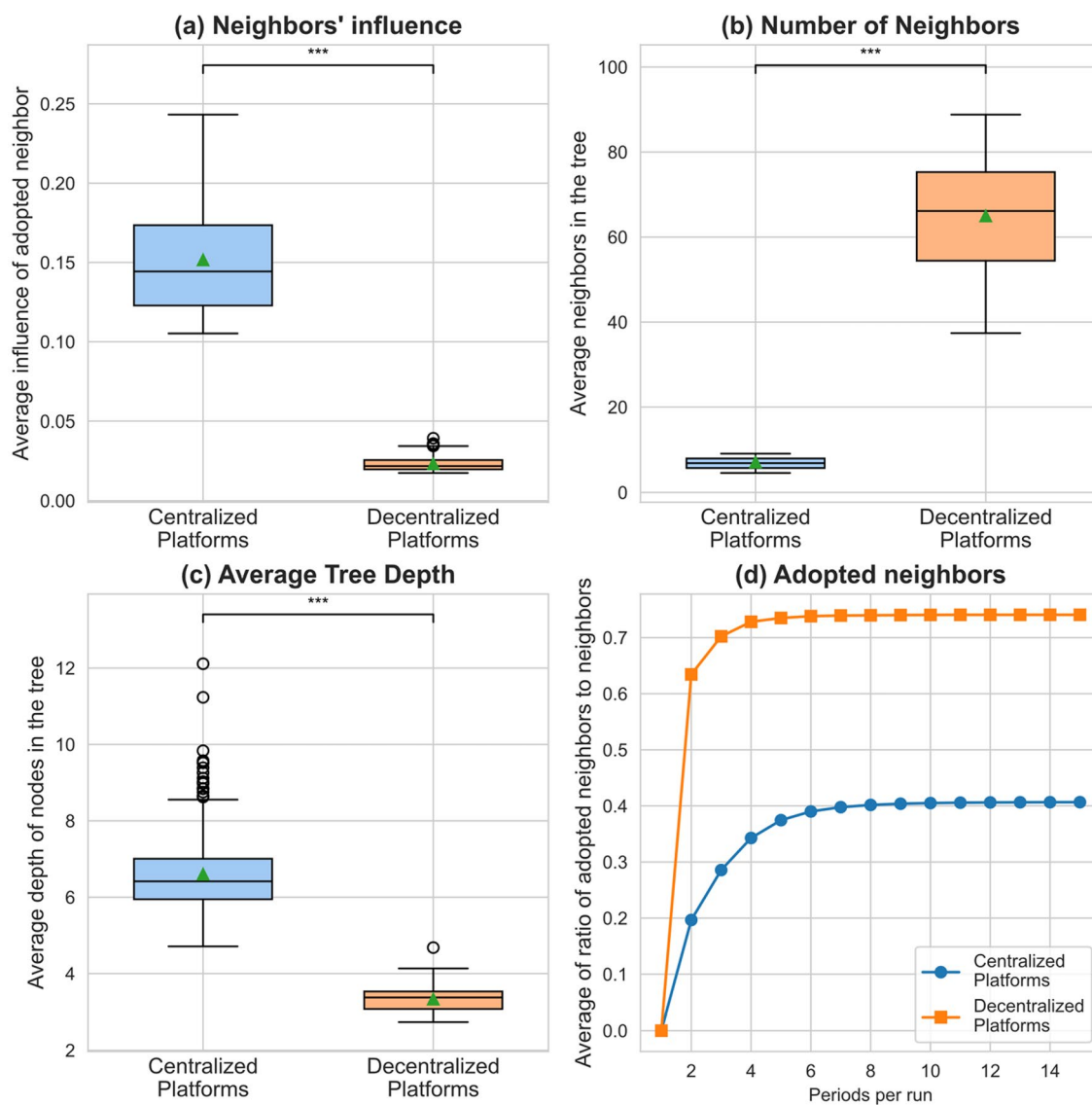


FIGURE A1 | Why are misinformation diffusion dynamics different between centralized and decentralized social network platforms? (a) shows that relative to the decentralized platform, the influence of neighbors in centralized platforms is greater because of fewer neighbors on average in centralized platforms, as shown in (b). The structural virality differences are driven by differences in average tree depth, as shown in (c). (d) explains differences in diffusion size that are explained by differences in adopted neighbors across centralized and decentralized platforms. While (a–c) aggregate values across 500 runs at the run level, (d) plots the average ratio of adopted users to neighbor users per period across 500 simulation runs. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Note that not all y-axis values start at zero.

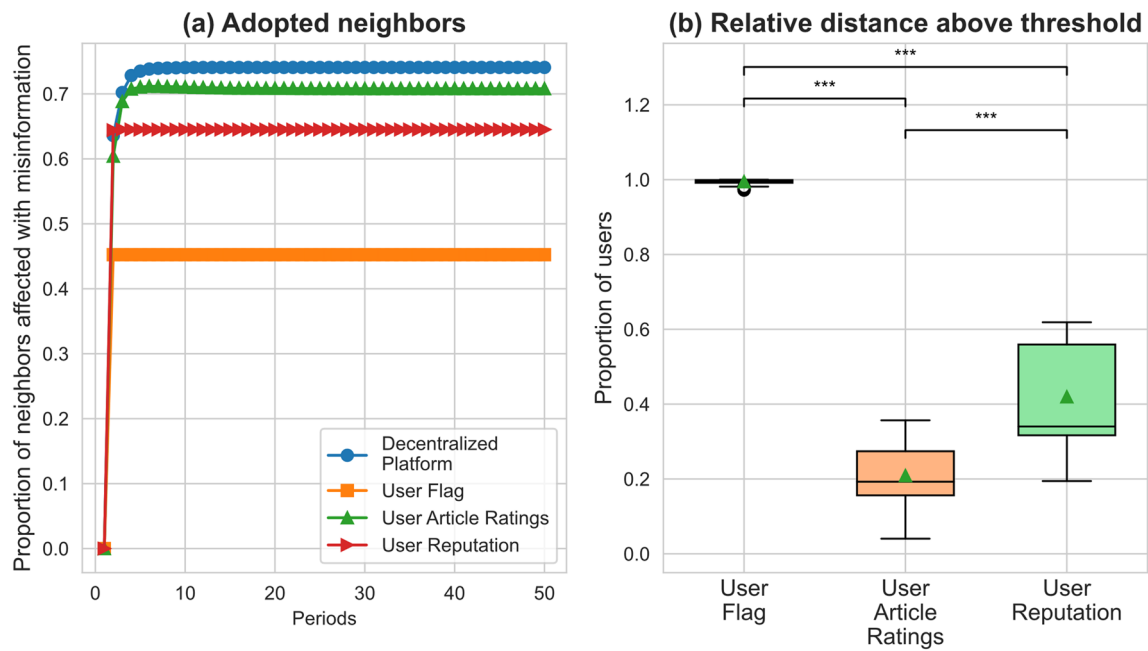


FIGURE A2 | Why are decentralized governance mechanisms for misinformation mitigation effective on decentralized social network platforms? Based on this figure, we conclude that the effect of the user flagging mechanism on structural virality and diffusion size can be attributed to overall low adoption rates; and the effect of the user reputation rating mechanism on global cascade can also be attributed to relatively low adoption rates. (a) plots the average proportion of adopted neighbors relative to all neighbors across systems in each period. In (b), each boxplot shows the minimum, first quartile, median, third quartile, maximum, and mean for the proportion of users above the threshold for each platform. Data for this plot are generated by 500 simulation runs, where each run has 50 periods. Wherever significant, each boxplot pair also shows the statistical significance using the Wilcoxon signed-rank test. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A1 | Related work.

Article	Platform	Research focus and theory	Method, data, and key variables	Key findings
Candogan and Drakopoulos (2020)	Centralized	Designing signaling mechanism to maximize engagement and minimize misinformation	– Analytical model – Simulation using Facebook subgraph	Signal engagement when inaccuracy is below a threshold; signal disengagement otherwise
Papanastasiou (2020)	Centralized	Optimal implementation of corrective measures (detecting and removing fake news in circulation) – Social learning	– Analytical model	Fake news proliferates in absence of intervention through sequential sharing. Understanding deferring of fact-checking to the agents results in gains (fact-checking costs and combating fake news)
Mostagir and Siderius (2023)	Centralized	Impact of information inequality on misinformation diffusion in a network of users	– Analytical model	Increase in misinformation in communities that lock in information
Mudambi et al. (2024)	Centralized	Efficacy of prominence reduction policy on misinformation diffusion	– Econometric models using data from Reddit platform	Prominence reduction policy diminishes misinformation contributions
Choi et al. (2020)	Decentralized	How blockchain technology support enhances social media analytics for supply chain and operations management	– Literature review – Review articles from SCOM journals on the use of social media for operational issues	Blockchain technology-supported social media can increase information diffusion efficiency
This paper	Decentralized	Misinformation diffusion and mitigation mechanisms for decentralized platforms – Theory of information diffusion – Complex adaptive systems theory – Community consensus	– Agent-based simulation model – Data from Pokec (centralized) and Steemit (decentralized) – Measures: structural virality, diffusion size, and global cascade – Mitigation mechanisms: user flagging, user article rating, and user reputation rating	Key findings: misinformation diffusion in decentralized platforms reach shorter distance from the misinformation initiating user, affects a larger number of users, and at a faster rate in comparison to centralized platforms; decentralized governance mechanisms for misinformation mitigation are effective in reducing the spread of misinformation on decentralized platforms

Note: A common theme across these OM studies is the focus on operational levers used by platforms to mitigate misinformation diffusion. This is one of the first studies to consider the mechanisms through which misinformation spreads in decentralized platforms and propose the use of community-based governance mechanisms for stopping the spread of misinformation on decentralized platforms.

TABLE A2 | Summary of agent-based simulation analyses.

Analysis	Description of analysis	Location
Part A: key results		
Misinformation diffusion in blockchain-based decentralized platform reach shorter distance from the misinformation initiating user, affects a larger number of users, and at a faster rate in comparison to centralized platforms	Empirical analysis suggests that relative to centralized platforms, misinformation diffusion in decentralized platforms has lower depth, larger number of connections, and greater number of connected neighbors	Manuscript Section 4.1 and Supporting Information: Ecompanion C
Commons-based misinformation mitigation mechanisms are effective in reducing the spread of misinformation on decentralized platforms	Commons-based misinformation mitigation mechanisms reduce the structural virality and size of misinformation diffusion and extend the time required to reach global cascade	Manuscript Section 4.2 and Supporting Information: Ecompanion C
Part B: key exploratory studies		
Empirical experiment to explain differences in misinformation diffusion dynamics between centralized and decentralized platforms	We consider neighbors' influence, average number of neighbors in a tree, average tree depth, and the proportion of adopted neighbors	Manuscript Section 4.1 and Figure A1
Empirically explain the efficacy of misinformation mitigation mechanisms in decentralized platforms	We compare differences in the proportion of adopted neighbors, and users whose mechanism-specific values are over the threshold	Manuscript Section 4.2 and Figure A2
Empirically demonstrate and explain the efficacy of misinformation mitigation mechanisms in centralized platforms	We compare differences in the proportion of adopted neighbors, and users whose mechanism-specific values are over the threshold	Supporting Information: Ecompanion C
Part C: key mechanism and robustness tests		
Evaluate misinformation diffusion dynamics on centralized (decentralized) platforms without the influence of a global controller (economic incentives)	We consider alternative systems in which the influence of a global controller and economic incentives is not modeled. Results are qualitatively the same	Supporting Information: Ecompanion C.3.2
Explore the aggregate effect of misinformation mitigation mechanisms	We consider an alternative system in which user flagging, user article rating, and user reputation rating are concurrently applicable. This combined system outperforms other mechanisms	Supporting Information: Ecompanion C.5.5
Evaluate dataset differences related to time frame (1999–2011 for Pokec and 2016–2019 for Steemit) and period (12 years vs. 3 years)	We consider another dataset from a centralized platform (Petster) to validate our results. Qualitative results remain the same	Supporting Information: Ecompanion C.5.6